UPDATE BRIEF II FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT* HISTORICAL ACCURACY OF THE KIRMSE–COXETER CORRECTION

JENS KÖPLINGER

Abstract. The manuscript version of 29 April 2026 (v3) is frozen while under review. This is the *second*, separate update brief for that frozen version; it is not to be merged with the first (`update_brief_2026-04-29_v3.tex`, a footnote correction and an appendix simplification). The present brief concerns one topic: how the manuscript describes the Kirmse–Coxeter correction of the integral octonions (the "Kirmse twist"). The four primary sources have now been obtained and verified from scratch in exact arithmetic; the brief is no longer provisional. It records what was verified and the resulting corrections, and supplies replacement text.

1. Verified from primary sources

Each source was read in the original and re-derived computationally; no secondary source was taken on trust. Verification notes are in `paper/reviews/`; scripts in `python_project/src/`.

- **Kirmse (1924).** He gives a correct octonion multiplication table and seeks the maximal orders. He states there are *eight* ("Man findet acht solche . . . ," p. 70) and exhibits one, J_1, in detail. J_1 *is not closed under multiplication*: the product $\frac{1}{2}(1+i_1+i_2+i_3)\cdot\frac{1}{2}(1+i_1+i_4+i_5) = \frac{1}{2}(i_1+i_2+i_4+i_7)$ leaves J_1. The true number of maximal orders is *seven*; J_1 is the spurious entry. (`verify_kirmse_1924.py`; note 2026-05-20.)
- **Dickson (1923).** Predating Coxeter, Dickson gave a genuine integral-octonion construction – his order O_D *is* closed. His Theorem XV nonetheless undercounts (states three maximal orders; there are seven), because of an extra assumption he states openly in the derivation: that the orders contain the Hurwitz quaternions. (`verify_dickson_1923.py`; note 2026-05-20.)
- **Mahler (1942).** Ideal theory over Coxeter's order; results correct, one misprint. (`verify_mahler_1942.py`; note 2026-05-20.)
- **Coxeter (1946).** Section 4 identifies Kirmse's error ("Actually there are only seven"), with the *same* non-closure witness as above. The remedy is due to **R. H. Bruck**: "Bruck's domain J can be derived from J_1 by transposing two of the i's." That is, a transposition of two *imaginary* octonion units – a linear involution of $\mathbb{R}^8$. The $\binom{7}{2} = 21$ such transpositions yield the seven maximal orders. Coxeter's postscript independently identifies the cause of Dickson's undercount (the integral-quaternion assumption). (`verify_coxeter_1946.py`; note 2026-05-22.)

2. Corrections to the manuscript

(a) **The correcting operation.** It is a transposition of two *imaginary coordinate axes* – a linear involution of $\mathbb{R}^8$, the same *kind* of map as the manuscript's σ. Any wording that describes it otherwise (e.g. as interchanging the identity with an imaginary unit, or as a Fano-triple permutation that is "not induced by a linear map") is to be removed.

Date: 22 May 2026.

(b) **Credit Bruck.** Coxeter *found* the error; Bruck *supplied* the remedy; Coxeter published and developed it. The manuscript credits only Coxeter (Remark in Section 2.3). Add Bruck.

(c) **Remark "σ vs. a Fano-line permutation" (Section 3).** It asserts that Coxeter's correction "rewrites the multiplication table without being induced by a linear map on $\mathbb{R}^8$." This is **false**: Coxeter's correction *is* a linear coordinate transposition. Replace the Remark with the text of Section 3.

(d) **Section 6 historical paragraph.** It carries the same false claim ("Coxeter's correction is a permutation of Fano *triples* ... without being induced by a linear map on $\mathbb{R}^8$"). The "different strata" framing there is correct and is kept; the "different *kind* of map" / "not a linear map" contrast is to be replaced by the accurate statement: the two are the same kind of map (a coordinate transposition), differing in the lattice each acts on – Coxeter's on a maximal-order candidate, the present σ on the sublattice Ls.

(e) **Remark on Kirmse's convention (Section 2.3).** It says "Kirmse originally proposed a specific labelling under which D_8^+ is not closed." This locates the error in the multiplication table; Coxeter's Section 4 – and our verification – locate it in the *lattice* Kirmse exhibited (J_1): his table is itself a correct octonion algebra. Re-word so the error is the choice of lattice, and so that Kirmse is recorded as stating *eight* maximal domains where there are seven.

(f) **Bibliography.** Add Dickson (1923); optionally Mahler (1942). A broader historical appendix (treating all four sources) is a separate deliverable; this brief covers only the wording of the twist.

3. Replacement text

The following replaces the Remark "σ vs. a Fano-line permutation."

> **Remark** (σ and the Coxeter correction). The Coxeter–Bruck correction of Kirmse's integral-octonion convention and the present σ are the same kind of map: a transposition of two imaginary coordinate axes, a linear involution of $\mathbb{R}^8$. Like any coordinate transposition, each rewrites the multiplication table – it does not preserve the Fano line structure – but each *is* induced by a linear map. The two differ only in what they act on. Coxeter's transposition repairs a maximal-order candidate that sits asymmetrically in $\mathbb{R}^8$. The present σ acts where the order is $L = D_8^+$, which is invariant under every coordinate permutation; hence $\sigma(L) = L$ and
>
> $$L \cdot_\sigma L \;=\; \sigma(L \cdot L) \;\subseteq\; L,$$
>
> so the twist preserves the E_8-level closure automatically. The twist's effect falls entirely on the sublattices Ls and $L\bar{s}$, which are not σ-invariant: that is what redirects Wilson's sublattice conditions.

The Section 6 paragraph is corrected to the same effect, keeping its "different strata" language: Coxeter's transposition and σ are one kind of operation, applied at different strata – Coxeter's to the E_8 order, σ to the condition-(3) sublattice.

Note on the companion document and reference files

`companion.tex` has been revised (22 May 2026); its Sections 4 and 10 now carry the corrected, example-driven account and a short proposition that $L \cdot_\sigma L \subseteq L$ for every coordinate transposition. The project reference files (`background_octonions.txt`, `terminology.md`) are still to be checked for the same imprecise wording and for a discrepancy in the name of Kirmse's journal.